

Paper 2 · Answers & Solutions

2026 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	A	B	C	B	B	C	D	C	B	A	D	D	A
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	B	A	D	C	B	B	C	C	D	C	D	A	D	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	B	A	B	C	C	D	A	A	D	A	A	C	D	C

Part 2 · Worked Solutions

Section A

1 A

B
Factorization

- $2m^2 - 7mn + 3n^2 = (m - 3n)(2m - n)$
- Take out $(m - 3n)$:
 $(m - 3n)(2m - n + 1)$, choose A

2 B

B-
Indices

- $2^{-1} + 2^{-3} = \frac{5}{8}$
- Reciprocal = $\frac{8}{5}$, choose B

3 A

B-
Linear Equations

- $\frac{a}{2} - \frac{a}{8} = \frac{3a}{8} = 9$
- $a = 24$, choose A

4 B

B
Approximation

- $\frac{1}{97} = 0.010309\dots$
- 3 significant figures = **0.0103**, choose B

5 C

B
Inequalities

- $5 - x < 3x - 3 \Rightarrow x > 2$
- $8 - 4x > 4 \Rightarrow x < 1$; union $x < 1$ or $x > 2$, choose C

6 B

B
Functions

- $f(3) - f(-3) = (27 - 12 + k) - (27 + 12)$
- $= -24$, choose B

7 B

B+
Remainder Theorem

- Divisible by
 $x - 7 \Rightarrow p(7) = 0 \Rightarrow c = -21$
- Remainder = $p(-\frac{1}{2}) = -15$, choose B

8 C

B+
Identities

- Compare coefficients of both sides
- Solving gives $m = 4$, $n = -8$, choose C

9 D

A-
Graphs of Quadratic Functions

- Vertex $x = -\frac{5}{p}$, $y = q$; from graph vertex in 3rd quadrant
- $p > 0$ and $q < 0$, choose D

10 C

B+
Compound Interest

- $A = 2500(1.03)^6$
- Interest = $A - 2500 \approx 485$, choose C

11 B

B+
Ratios (Maps)

- Area ratio = $1 : 25000^2$
- $= 1.875 \times 10^5 \text{ m}^2$, choose B

12 A

B+
Partial Variation

- $y = c + kx^2$: $c = 5$, $k = 4$
- $x = 5$: $y = 105$, choose A

13 D

B+
Sequences (Patterns)

- $a_n = 1 + (n - 1)(n + 2)$
- $a_7 = 1 + 54 = 55$, choose D

14 D

A-
Triangles (Pythagoras)

- $6AD - 6DC = 24 \Rightarrow AD - DC = 4$,
 $AD + DC = 14$
- $AD = 9$, $DC = 5$; perimeter
 $= 15 + 13 + 14 = 42$, choose D

15 A

A-
Volume of Solids

- Cone $\frac{1}{3}\pi(2r)^2h = 36\pi \Rightarrow r^2h = 27$
- Cylinder = $\pi r^2(3h) = 81\pi$, choose A

16 D

A
Ratios of Areas

- Compute area ratio via affine coordinates
- $[DFGH] : [ABG] = 16 : 15 \Rightarrow 144$, choose D

17 B

A
Similar Triangles

- Using $\angle CDE = 60^\circ = \angle B$ and the equilateral triangle
- $CE = 14$, choose B

18 A

A
Isosceles Triangles

- $AB = BC$, $AC = AD$, $AE \parallel BC$, marked 124°
- $\angle BCA = 56^\circ$, choose A

19 D

A-
Pythagoras (Grid)

- Horizontal 8, vertical 6
- $AH = \sqrt{64 + 36} = 10$, choose D

20 C

A
Parallelograms

- E mid-point of AD ,
 $\angle ABE = \angle CBD = \angle DBE$
- I and II hold, III not necessarily, choose C

21 B

A
Properties of Circles

- $\angle ABC = 110^\circ$; AD diameter, $BC = CD$
- arc $BCD = 80^\circ$, $\angle BED = 40^\circ$, choose B

22 B

A
Trigonometry

- $CE = BC \tan 40^\circ$
- $\angle AED \approx 57^\circ$, choose B

23 C

A
Graphs of Straight Lines

- Compare slopes and intercepts of L_1, L_2
- I and II hold, choose C

24 C

A-
Equations of Straight Lines

- Given slope $\frac{8}{3}$, L slope $= -\frac{3}{8}$
- Through $(-4, 0)$: $3x + 8y + 12 = 0$, choose C

25 D

A
Polar Coordinates

- Convert to rectangular; distance from Q to line PR
- Perpendicular distance $= 7$, choose D

26 C

A
Equations of Circles

- C_2 : centre $(-2, 1)$
- G_1, G_2, O collinear; radii & distances unequal; only I, choose C

27 D

A
Locus

- $AP = BP \Rightarrow P$ on perp. bisector through centre $(3, 2)$
- $3 + 4 + k = 0 \Rightarrow k = -7$, choose D

28 A

A-
Probability (Bar Chart)

- Fewer than 5: frequency $4 + 8 = 12$, total 30
- $P = \frac{2}{5}$, choose A

29 D

B+
Box-and-Whisker Diagrams

- Left edge of box is the lower quartile
- Lower quartile $= 15$, choose D

30 B

A
Statistical Measures

- $m + n = 5$; max 10, min 1
- Range must be 9; only III, choose B

Section B

31

A

A
Transformations of Graphs

- 1 $g(x) = f(\frac{x}{2})$ is a horizontal stretch by factor **2**
- 2 Choose the horizontally-stretched Graph A, choose A

32

B

A
Number Representation

- 1 In base 8, 8^{12} is **1** followed by twelve **0**s; 8^3 is a **1** at position **3**
- 2 $= 1\,000\,000\,001\,000_8$, choose B

33

A

A
Logarithms (Linear Relation)

- 1 Slope $= 1 = \log_2 b$, intercept $\log_2 a = 3$
- 2 $y = 8 \cdot 2^x$, choose A

34

B

A
Exponential Equations

- 1 Let $t = 3^x$: $t^2 - 4t + 3 = 0 \Rightarrow t = 1$ or **3**
- 2 $x = 0$ or **1**, choose B

35

C

A
Complex Numbers

- 1 $\frac{5-2i}{k-i}$ real $\Rightarrow$ imaginary part $= 0$
- 2 $5 - 2k = 0 \Rightarrow k = \frac{5}{2}$, choose C

36

C

A
Sequences

- 1 I: $\lg 2, 2\lg 2, 3\lg 2$ arithmetic
- 2 II geometric; III arithmetic; I and III, choose C

37

D

A
Linear Programming

- 1 Evaluate $x - 2y + 43$ at vertices
- 2 Max at **(9, 0): 52**, choose D

38

A

A
Rectangles (Coordinates)

- 1 $AC = 35, AE = 30$ on AC
- 2 $DE = \sqrt{585} = 3\sqrt{65}$, choose A

39

A

A
Angles in 3-D

- 1 $BD = 20$; distance B to CD by Heron
- 2 $\sin \theta = \frac{20}{25} \Rightarrow 53^\circ$, choose A

40

D

A
Circles and Tangents

- 1 Tangent-chord $\angle ACB = 68^\circ$; $\angle BCO = 26^\circ$
- 2 Using $OC = OB, \angle ABC = 48^\circ$, choose D

41

A

A+
Incentre of Triangle

- 1 In-centre $(r, r), r = \frac{p+q-\sqrt{p^2+q^2}}{2}$
- 2 $7r = 3p \Rightarrow p : q = 7 : 24$, choose A

42

A

A
Combinations

- 1 $C_5^{13} \times C_4^6$
- 2 $= 1287 \times 15 = 19305$, choose A

43

C

A-
Binomial Probability

- 1 At most 3 hits $= 1 - 0.6^4$
- 2 $= 0.8704$, choose C

44

D

A-
Standard Scores

- 1 $-2.5 = \frac{35 - 50}{sd}$
- 2 $sd = 6$, choose D

45

C

A-
Transformations of Statistics

- 1 Each $\times 8$: mode $\times 8$, IQR $\times 8$
- 2 Variance $\times 64$; I and II only, choose C

